

Matthew K. McGrath

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SUMMARY

Software engineer with 4+ years of daily hands-on UNIX/Linux experience across Arch, Debian, and Gentoo, plus high-volume semiconductor fab experience at Intel Ocotillo. Designs, deploys, and operates multi-node Linux infrastructure end to end — systemd services, /proc-level instrumentation, shell tooling, networking, and firewalling.

TECHNICAL SKILLS

- **UNIX / Linux:** 4+ years daily use across Arch, Void, and Debian; systemd unit authoring and service management, /proc and /sys instrumentation, Bash scripting, permissions and package management, log analysis, and shell-level troubleshooting of boot, init, storage, and networking issues
- **Systems & Networking:** SSH key authentication and hardening, UFW firewall configuration, static network topology and TCP/IP fundamentals, Docker and Docker Compose, Traefik reverse proxying, DNS filtering with Pi-hole
- **Programming & Application:** Python, C, Bash, C#, JavaScript, SQL; FastAPI, .NET, REST API design, PostgreSQL, Git, React, React Native, terminal UI (TUI) development
- **Fab Operations:** Automated material handling (PGV), cleanroom protocol, nonstandard event escalation

TECHNICAL PROJECTS

Sentinel — Distributed UNIX Monitoring System

January 2026 – July 2026

Personal Project — github.com/mkmcgrath/sentinel

Mesa, AZ

- Architected and deployed a distributed UNIX monitoring platform spanning 3 physical Linux nodes, continuously collecting CPU, memory, disk, and network telemetry to surface degradation before failure
- Engineered a lightweight Python agent running as a systemd service on each node, reading metrics directly from the Linux /proc filesystem with zero external dependencies for reliability on constrained hardware
- Built a FastAPI/PostgreSQL backend exposing a REST API, with both a terminal UI (TUI) and a web dashboard
- Configured threshold-based alerting to convert reactive troubleshooting into proactive intervention
- Owned the full UNIX operations lifecycle: Docker Compose orchestration, SSH hardening, UFW firewall rules, and static network topology across all nodes

Self-Hosted Linux Service Infrastructure

2024 – Present

Personal

Mesa, AZ

- Administer a multi-node self-hosted Linux stack (Nextcloud, Jellyfin, Pi-hole, Traefik) in continuous daily use, handling upgrades, service recovery, and configuration drift
- Maintain reverse-proxy routing through Traefik, DNS-level filtering via Pi-hole, services containerized via Docker Compose

PROFESSIONAL EXPERIENCE

Full Stack Engineering Intern

January 2026 – June 2026

Meritus360

Remote

- Designed and deployed a Slack-integrated uptime monitoring bot for critical systems, cutting detection-to-response time from hours to ~5 minutes via real-time outage alerts
- Developed full-stack application features using React, .NET (C#) REST APIs, and PostgreSQL

Manufacturing Operator (PGV)

June 2025 – October 2025

Kelly Services @ Intel Ocotillo (OTF)

Chandler, AZ

- Operated automated material handling (PGV) systems in a high-volume semiconductor fab, coordinating WIP movement across the floor to sustain throughput targets
- Worked full cleanroom gowning protocol on compressed 12-hour shifts under strict contamination-control and standard operating procedures, executing structured shift-change handoffs

Full Stack Engineering Intern

January 2023 – August 2023

The Wolff Company

Scottsdale, AZ

- Built and maintained 5+ internal application features using .NET and React Native; participated in sprint planning and weekly engineering progress reviews

EDUCATION

B.S. Software Engineering

July 2021 – December 2026

Brigham Young University – Idaho — completing coursework while employed full-time

Rexburg, ID (Remote)